

01 The corrected concept

Seismic Eddy Current and Active Damping Response

An educational model that connects a moving ground base, four spring supports, a building platform, a roof-mounted motion sensor and two independently switched solenoids. This revision follows the clarified mechanical layout and the relay-based circuit supplied with the project.

What is fixed, and what moves?

Part	Arrangement and purpose
Ground base	The earthquake input. Four corner bolts are fixed to this base. There are no separate tall pillars.
Four springs	One longer spring at each corner, seated over a base-fixed screw with 1/2 inch (12.7 mm) inserted into the spring. Together they carry the upper platform.
Building platform	Rests on all four springs and carries the model building. It can move relative to the base.
Solenoid 01	Between spring S1 and S2. Centered compact push-pull unit; illustrated plunger moves left toward S1 on energization.
Solenoid 02	Between spring S3 and S4. Centered mirrored push-pull unit; illustrated plunger moves right toward S4 on energization.
MPU6050	Fixed to the roof of the building. Reads building acceleration and angular motion.

The essential correction

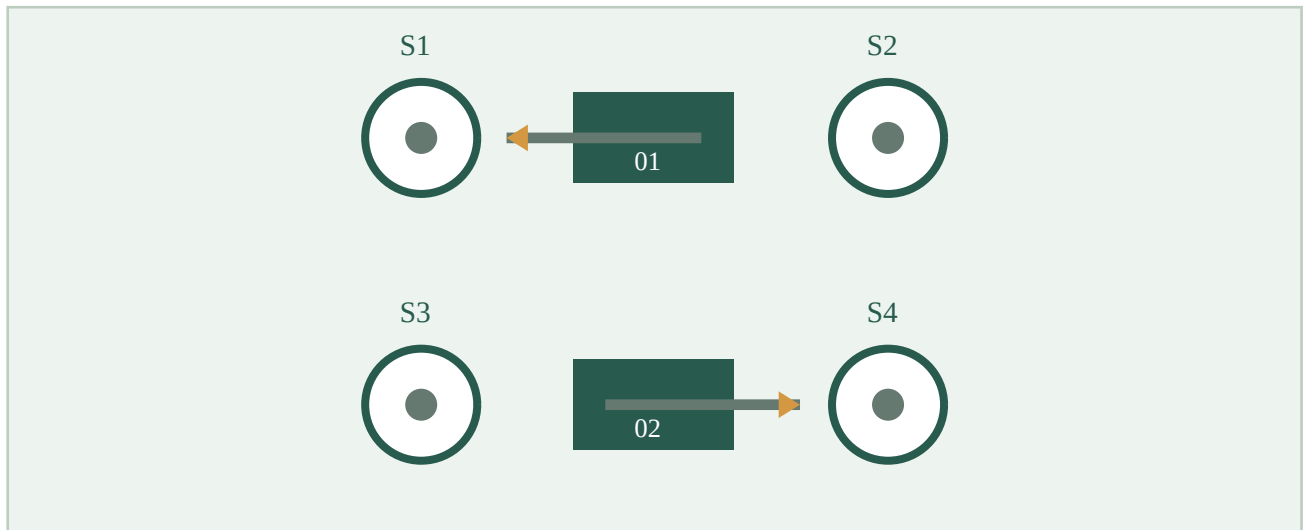
The solenoid rods act on the **springs**, not directly on the building. The spring reaction is transmitted through the spring seats into the moving platform. The previous eight-spring illustration and freestanding side pillars are superseded.

Confirmed by the project owner: four springs, a base-fixed screw inserted 1/2 inch into each longer spring, two mirrored compact push-pull solenoids between the spring pairs, a platform above the springs and the sensor on the roof.

Proposed visualization details: base-fixed coil brackets and small spring-contact collars. These communicate a possible force path; they are not a finalized fabrication drawing. The 1/2-inch measurement specifies insertion depth, not screw diameter.

02 Mechanical arrangement

Top view, with the building and upper platform removed. S1-S4 identify the four corner springs; they are not additional pillars.



Assembly from bottom to top

1. Ground base, with a fixed threaded fastener at each corner.
2. Washer / retained lower spring seat around each screw; the screw enters the longer spring by 1/2 inch (12.7 mm).
3. Four springs and retained upper seats under the moving platform.
4. Upper platform, building, then roof-mounted sensor.

Bolt retention, spring end retention and clearance must allow the intended lateral movement without a bolt locking the platform rigidly to the base. The model exaggerates gaps so the mechanism can be inspected; displayed dimensions are not a scale drawing.

Solenoid mounting and contact

In the proposed visualization, each coil body is held to the ground base by a low saddle bracket. The short plunger protrudes from the inward side at rest. Energization slides it through the coil toward the opposite, spring-facing side. The second actuator is mirrored. The rod is not long enough to span the spring pair. A collar at the intended spring contact represents a distributed contact point rather than a sharp rod pressing on a single coil turn.

The supplied photo guides the compact frame, blue winding, plunger and return spring. The coil is centered; springs are closer and the illustrated plunger slightly longer. Fit real spacing to the actual stroke. A rigid plunger translates as one piece: one end advances while the opposite retracts. Verify stroke, force, contact height and alignment on the actual unit before fabrication.

03 Sensing and feedback

Why the sensor is on the roof

The roof position makes building sway and rotation observable. Mount the board rigidly to the model, align and record its axes, and leave a flexible wire loop so the cable does not become an unintended spring. The MPU6050 combines a three-axis accelerometer and a three-axis gyroscope [1].

A roof sensor measures the building response, not the ground motion independently. Acceleration also contains gravity effects as the structure tilts. It does not directly report displacement. Filtering, bias calibration and an appropriate motion estimate are needed; simply integrating acceleration produces drift.

Proposed operating sequence

Stage	What happens
1 / Initialize	Power the control circuit, check sensor communication and establish a stationary baseline. Keep both outputs inactive during startup or sensor failure.
2 / Detect	Read acceleration and angular rate. Filter noise and compare motion against experimentally selected thresholds.
3 / Estimate	Estimate the relevant motion direction. Acceleration sign alone is not the same as velocity direction.
4 / Select	For the illustrated X direction: a leftward spring force comes from channel 1; a rightward force comes from channel 2. Confirm this sign on the actual mechanism.
5 / Actuate	The selected relay closes COM to NO, energizing a 12 V solenoid. Its working rod acts on a spring contact.
6 / Observe	Measure the resulting response, release the actuator as required, and repeat. Use hysteresis and minimum switching intervals to avoid chatter.

Critical distinction: reducing vibration requires an opposing force at the right time. Incorrect timing or mounting can add energy. The web controller uses sampled model velocity, short pulses and a quiet zone; the hardware estimator and control must be developed and tested.

The roof sensor can observe motion about multiple axes. Two push-pull actuators both aligned along X do not provide independent active control of Y and Z. Their offset positions introduce twisting, now included as rigid-platform yaw. Flexible floors and vertical motion are not modeled.

04 Circuit and power paths

The current design uses a two-channel relay module, not an IRF520 module. Digital output pins D8 and D9 are the project assignments. The classic Nano uses A4 for SDA and A5 for SCL [2].

Connection	Destination / function
Adapter +12 V	Main switch, then buck IN+ and both relay COM terminals.
Adapter negative	Common ground distribution point.
LM2596 output	Adjusted regulated 5 V to Nano 5V and relay VCC; output negative to ground.
MPU6050 signals	SDA to A4; SCL to A5; optional INT to D2. AD0 to ground for the default address, when appropriate.
Sensor supply	Use the voltage and logic levels required by the exact breakout. A module regulator does not automatically guarantee 5 V-safe I2C pins.
Relay inputs	IN1 to D8, IN2 to D9. Verify active-LOW or active-HIGH behavior.
Relay channel 1	COM to switched 12 V; NO to solenoid 01 positive. NC unused.
Relay channel 2	COM to switched 12 V; NO to solenoid 02 positive. NC unused.
Solenoid negatives	Return directly to the common ground distribution point.
Flyback diodes	One across each solenoid: cathode / striped end to positive, anode to negative. Select for the actual coil and switching requirements.

Keep high-current coil returns separate from the sensor and Nano return paths until the common distribution point. Verify the buck output with a meter before connecting electronics. Check relay DC inductive-load suitability and supply current against the actual solenoid specifications.

The exact sensor breakout, coil current, duty cycle, stroke, force rating, spring rate, model mass and bolt dimensions have not been supplied. This reference intentionally does not invent those values.

05 Component responsibilities

Component	Quantity	Role
Arduino Nano	1	Reads the sensor and chooses relay states.
MPU6050 breakout	1	Reports acceleration and angular rate from the roof.
LM2596 module	1	Steps the 12 V source down to the regulated control supply.
12 V adapter + switch	1 each	Supplies and isolates the prototype power. Current rating must match the load.
Two-channel relay module	1	Independently switches each solenoid power path.
Push-pull solenoids	2	Apply the proposed leftward and rightward spring actions.
Flyback diodes	2	Provide a coil-current path when each solenoid is switched off.
Springs + corner anchors	4 each	Support the upper platform with compliance at all four corners.
Base, platform, building	1 each	Form the ground input and moving mechanical assembly.
Seats, brackets, contacts	As required	Retain springs and actuators while preserving intended movement.

Active actuation versus eddy-current damping

The project name is retained, but the current arrangement illustrates **active solenoid response**. A powered solenoid moving a rod is not by itself proof of eddy-current damping. The original PDF described a different arrangement with a moving conductive plate and a magnetic field.

A separate eddy-current stage would need a specified conductor, magnetic field geometry and relative motion. It is not included in this four-spring model. Springs store and return energy; energy reduction depends on dissipative effects and correctly timed control, not on springs alone.

Why the relay model is simplified

Real relays have finite switching delays and contact life. The revised web model switches relay states with delay and pulse limits rather than imposing an instant damping force. Coil inductance, flyback decay, contact gaps, rod travel and return forces also affect timing. Establish a modest, measured demonstration first rather than treating ideal simulation results as hardware performance.

06 Using and testing the model

The interactive website

Drag the 3D model to orbit; scroll or pinch to zoom; right-drag or use two fingers to pan. Front, Side, Top and Underside buttons provide repeatable views. Top and Underside reveal the mechanism by hiding the building and upper platform. Use Show building to reassemble the view.

Click the building or Trigger earthquake for ground excitation. Change intensity, compare active damping on and off, pause, or reset. The reference model at the bottom rotates independently and can be stopped. The guide uses Previous / Next navigation with a unique placement or wiring image for each step. Phone input requires a motion-capable browser, permission where requested, and HTTPS.

What the graphs mean

The model integrates lateral translation, yaw and two electromechanical plungers in SI units using RK4 at 2,000 steps per second. Four effective lateral springs oppose relative displacement. The base acceleration enters as an inertial load. Coil current, plunger inertia, return springs, unilateral contacts, stroke stops and relay delay determine the actual transferred force. A matched unpowered model receives the same initial conditions and base input.

The graphs show absolute displacement in millimeters; the intensity is a synthetic input, not a Richter magnitude. Reduction is signed RMS change: negative means worse motion. Example parameters are editable and are not hardware measurements. Contact uses a penalty stiffness and a linear collar-height force transfer approximation. Magnetic saturation, heating, spring-coil details, floor bending and vertical dynamics are excluded.

Suggested prototype validation

Check	Record and compare
Unpowered mechanism	Verify that four springs carry the load and that the platform can move without binding, detaching or reaching unsafe travel.
Actuator direction	At low excitation, verify which spring the short working end contacts and the resulting platform direction. Check stroke and clearance.
Sensor baseline	Log stationary bias and small known movements. Record mounting axes and sampling interval.
Passive run	Use a repeatable ground input with the actuators disabled; measure peak response and settling time.
Controlled run	Repeat the same input and initial conditions with control enabled. Compare traces, contact timing and coil temperature.

Check	Record and compare
Tuning	Adjust thresholds and timing gradually. Reject settings that increase motion or introduce twisting.

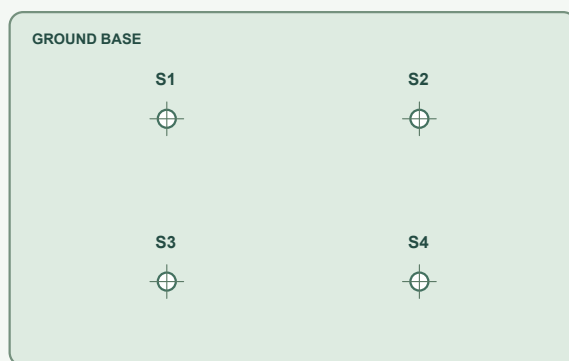
This is a college-scale experimental prototype, not a validated earthquake-protection system for occupied buildings.

Step 01 / Check the actual parts

Build the mechanism with all power disconnected. The 3D view shows arrangement, not fabrication dimensions.

01 / Check the actual parts

PLACEMENT VIEW / schematic, not to scale



Inventory and positioning

LAY OUT BEFORE DRILLING

- 4 matching springs
- 4 retained corner screws
- 2 mirrored push-pull coils
- Base + upper platform
- Building + roof sensor

Set spacing from actual stroke.

What to do

Record the solenoid voltage, current, stroke, force, return action and duty cycle; spring free length and rate; platform mass; relay input logic and DC inductive contact rating; and the exact sensor breakout voltage requirements. Use four matching springs. Choose an adapter and wiring rated for the combined load.

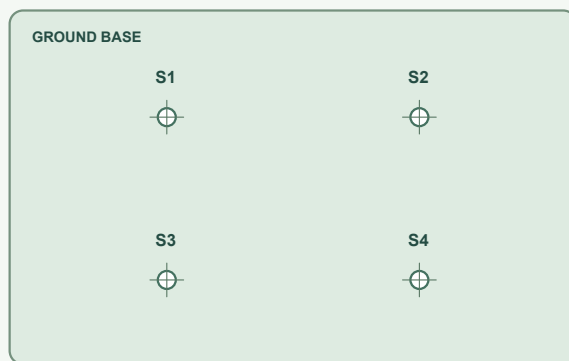
Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 02 / Lay out four spring centers

Build the mechanism with all power disconnected. The 3D view shows arrangement, not fabrication dimensions.

02 / Lay out four spring centers

PLACEMENT VIEW / schematic, not to scale



Four marked screw centers

LAY OUT BEFORE DRILLING

- 4 matching springs
- 4 retained corner screws
- 2 mirrored push-pull coils
- Base + upper platform
- Building + roof sensor
- Set spacing from actual stroke.

What to do

Mark S1 and S2 as one row, S3 and S4 as the other. Center one solenoid between each pair. Keep both rows parallel. Reduce spring spacing until the real plunger can reach its intended spring contact within its rated stroke, while retaining clearance for the opposite end. The slightly longer rod in the illustration is not an instruction to extend a real plunger.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 03 / Fit the base screws and longer springs

Build the mechanism with all power disconnected. The 3D view shows arrangement, not fabrication dimensions.

03 / Fit the base screws and longer springs

PLACEMENT VIEW / schematic, not to scale



What to do

Fix one screw securely to the base at each corner. Use a retained lower spring seat and suitable washer/nut arrangement. Each screw must enter its spring by 1/2 inch (12.7 mm), measured from the lower end of the spring. Select screw diameter to clear the spring bore; total screw length also includes base thickness and fastening hardware. Leave the rest of the spring free to flex.

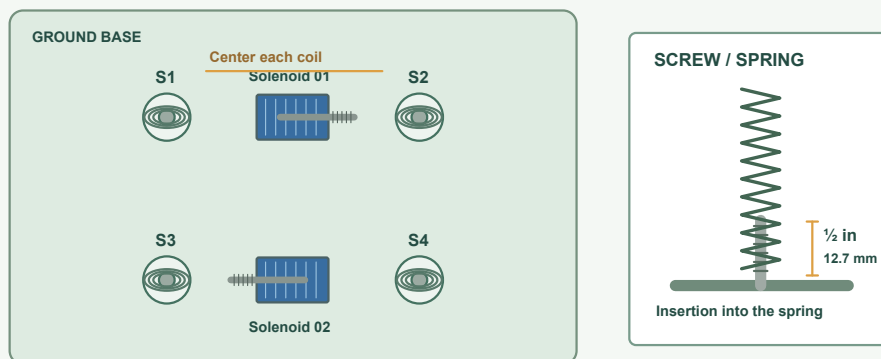
Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 04 / Seat the platform and mount the coils

Build the mechanism with all power disconnected. The 3D view shows arrangement, not fabrication dimensions.

04 / Seat the platform and mount the coils

PLACEMENT VIEW / schematic, not to scale



Centered coils; mirrored working stroke

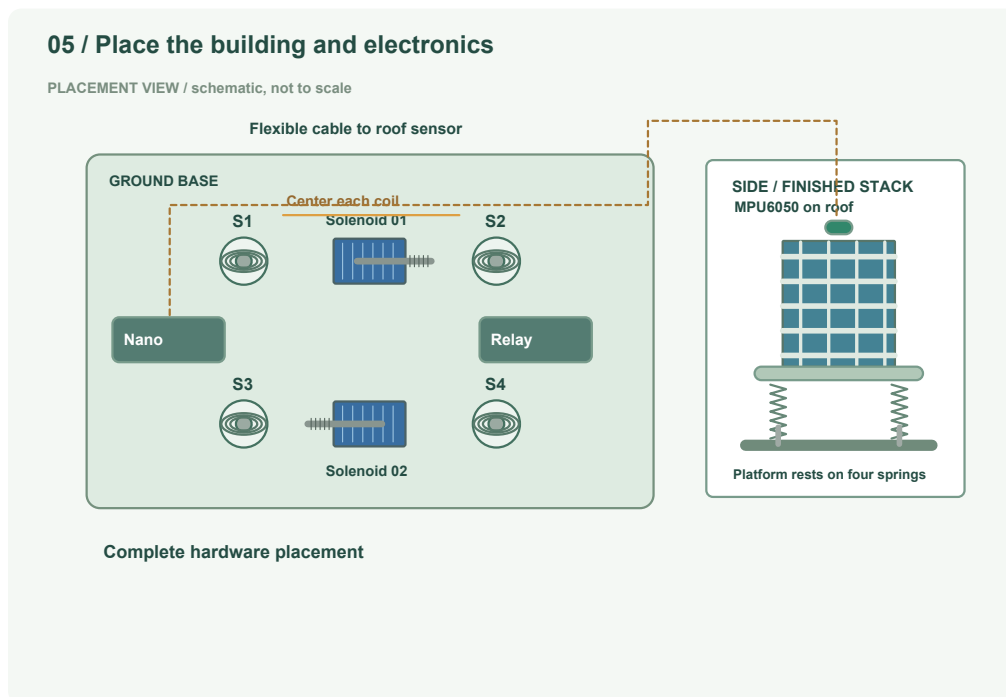
What to do

Retain the four upper spring ends under the platform so they cannot slip out during the demonstration. The corner screws must not clamp the upper platform rigidly to the ground base. Fasten the coil bodies to low brackets on the base, centered within the spring pairs. Mirror the units: channel 1's energized stroke points left, channel 2's points right. Align the short plunger with a spring-contact collar; the contact design is provisional and needs a bench check.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 05 / Place the building and electronics

Build the mechanism with all power disconnected. The 3D view shows arrangement, not fabrication dimensions.



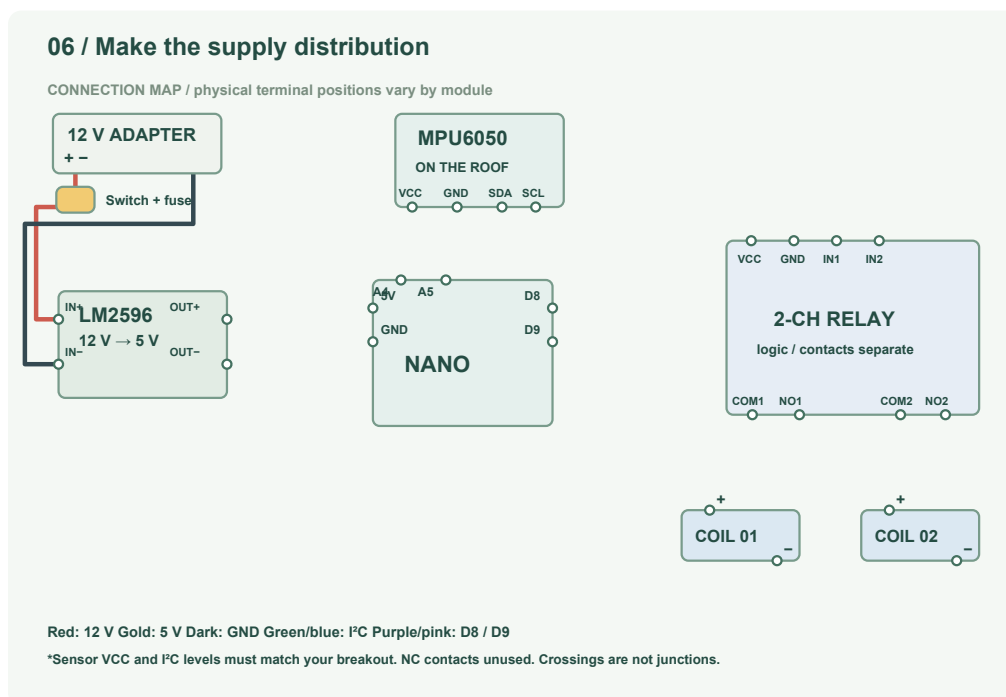
What to do

Secure the building centrally on the platform. Rigidly mount the MPU6050 on the roof and mark its axes. Keep the Nano, buck converter, relay module and power distribution on an insulated, accessible area of the base outside the moving mechanism. Route a flexible sensor cable loop to the roof. Move the platform by hand to verify clearance, retained spring ends and no binding.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 06 / Make the supply distribution

Keep the solenoid power branch disconnected during these steps. Use the component terminal labels, not wire colors alone.



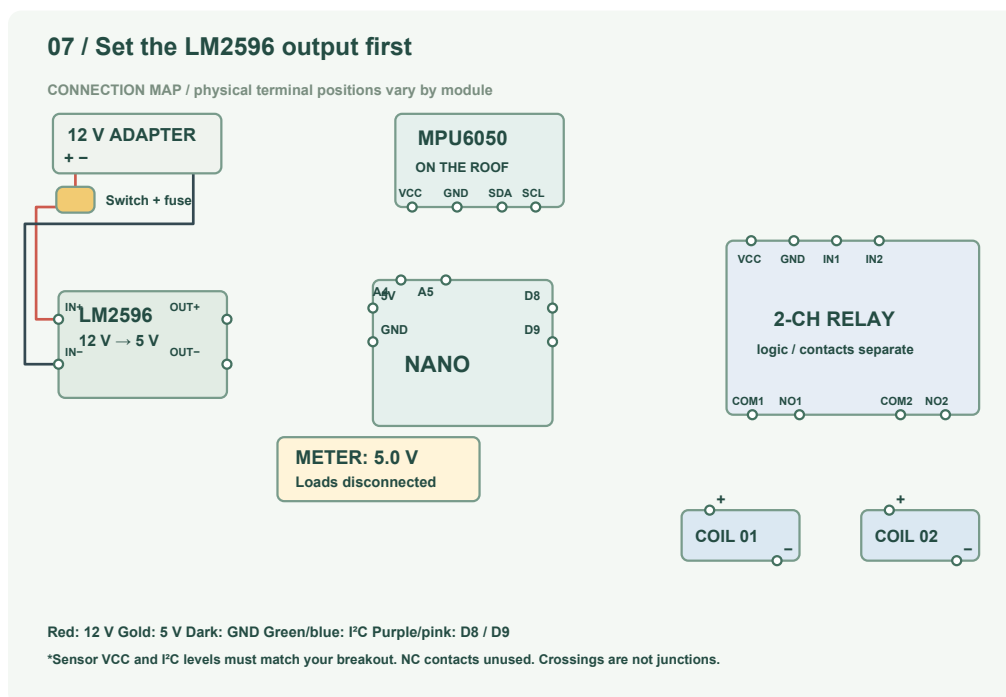
What to do

With the adapter unplugged, route its positive output through the main switch and an appropriately rated fuse to the switched 12 V distribution point. Route adapter negative to a separate common-ground terminal. Select fuse and cable ratings from actual load specifications; do not route coil current through a solderless breadboard or Nano pins.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 07 / Set the LM2596 output first

Keep the solenoid power branch disconnected during these steps. Use the component terminal labels, not wire colors alone.



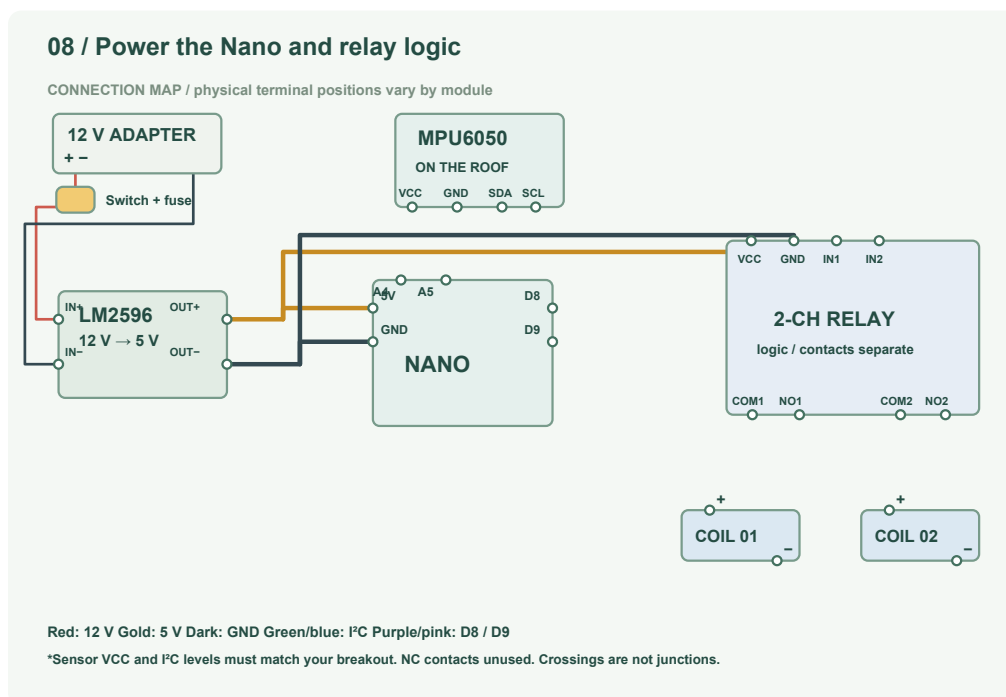
What to do

Connect buck IN+ to switched 12 V and IN- to common ground. Leave all output loads disconnected. Power only the buck, measure OUT+ to OUT- with a multimeter, and adjust to 5.0 V. Switch off and unplug the adapter before making the next connections.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 08 / Power the Nano and relay logic

Keep the solenoid power branch disconnected during these steps. Use the component terminal labels, not wire colors alone.



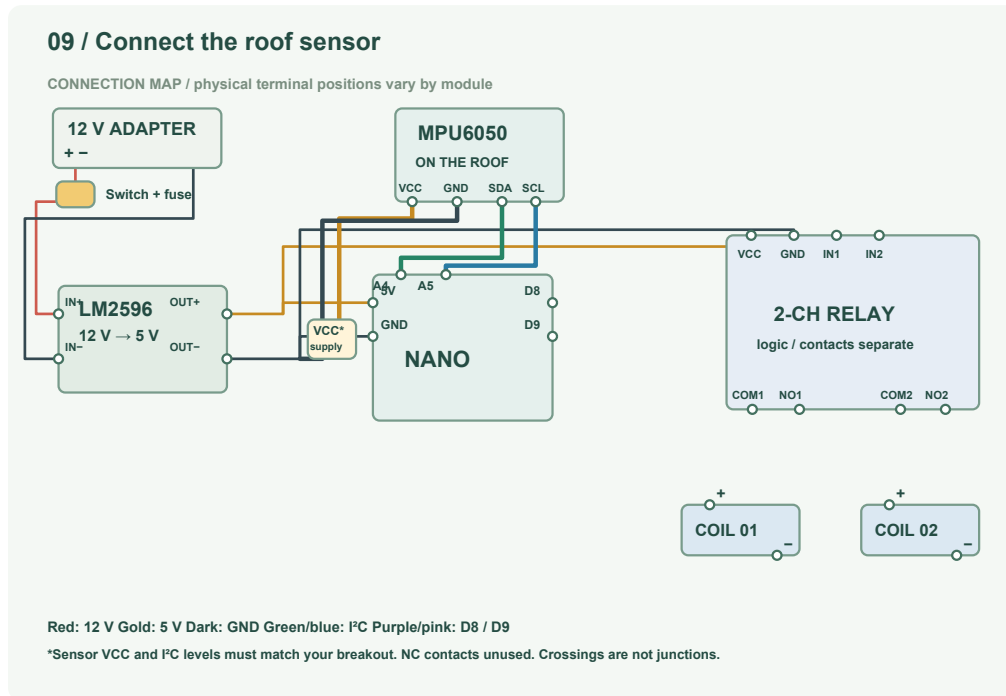
What to do

Connect regulated OUT+ to the classic Nano 5V pin and relay VCC; connect OUT- to their GND terminals. Do not feed regulated 5 V into VIN. This guide assumes a relay module with onboard drivers and a documented VCC/GND logic connection. Follow the particular module's JD-VCC/isolation instructions if its arrangement differs. Avoid simultaneous USB and external power unless the board's power arrangement is verified.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 09 / Connect the roof sensor

Keep the solenoid power branch disconnected during these steps. Use the component terminal labels, not wire colors alone.



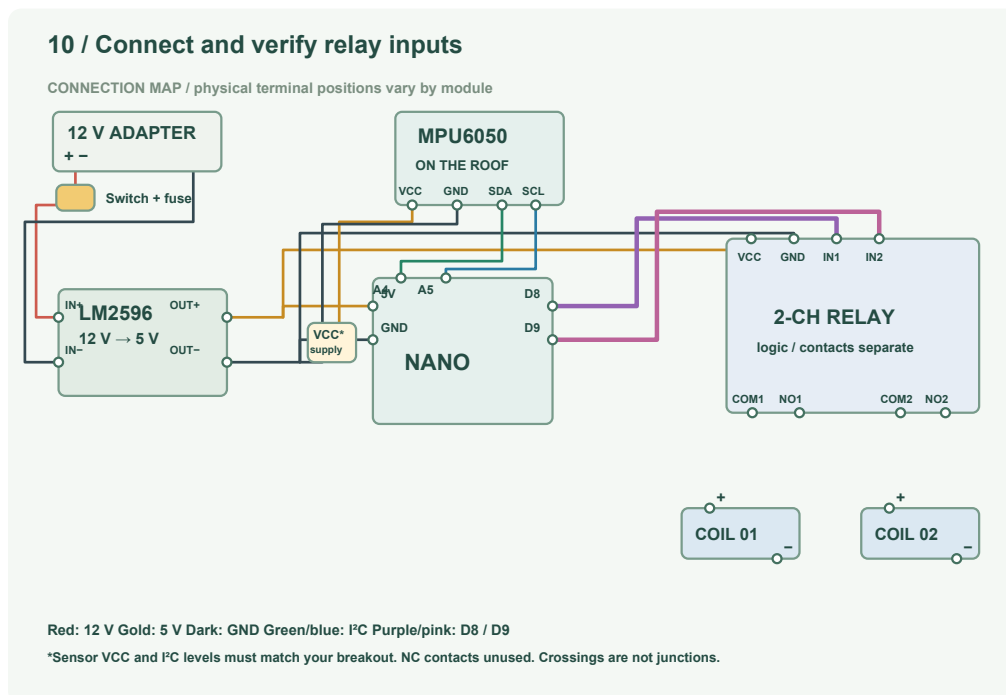
What to do

Power the MPU6050 breakout only at its documented voltage; use a suitable 3.3 V source or level translation if required. A regulator on a breakout does not establish I2C logic tolerance. Connect sensor GND to common ground, SDA to Nano A4 and SCL to A5. Optional INT goes to D2. AD0 may connect to GND for address 0x68. Leave XDA and XCL unused. Keep this cable away from coil wiring.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 10 / Connect and verify relay inputs

Keep the solenoid power branch disconnected during these steps. Use the component terminal labels, not wire colors alone.



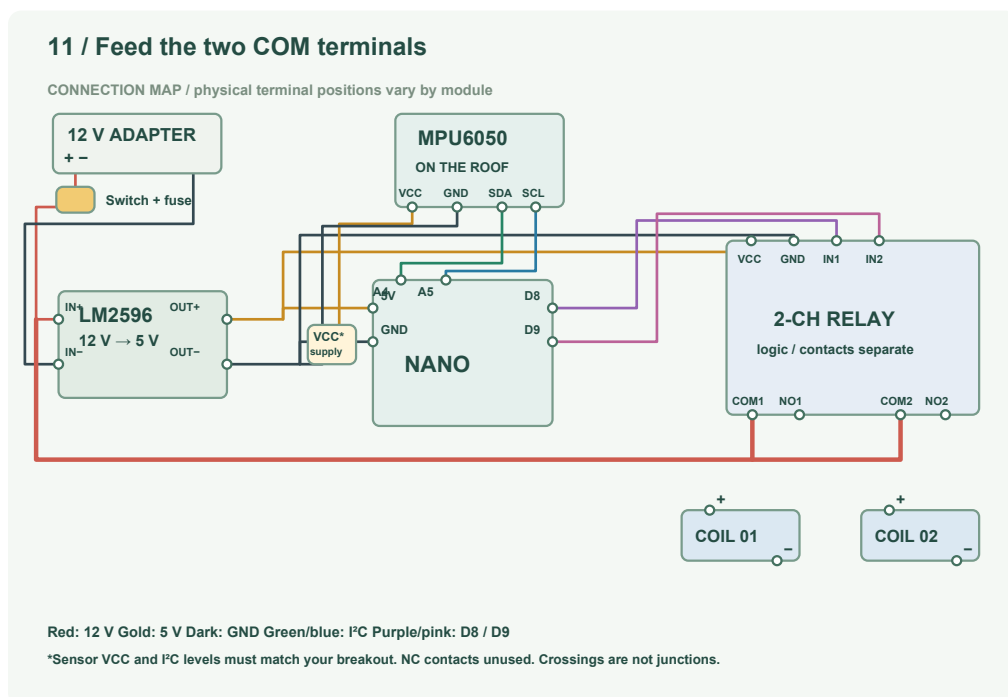
What to do

Connect Nano D8 to IN1 and D9 to IN2. Verify whether the relay board is active-LOW or active-HIGH using its documentation. With the 12 V contact branch still disconnected, test each relay independently with a simple known test sketch. Both channels must default OFF at startup and on a sensor fault. Firmware is not supplied by the frontend and must match the actual module.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 11 / Feed the two COM terminals

Disconnect all power before wiring. Relay contact terminals COM/NO/NC are separate from the module's logic pins.



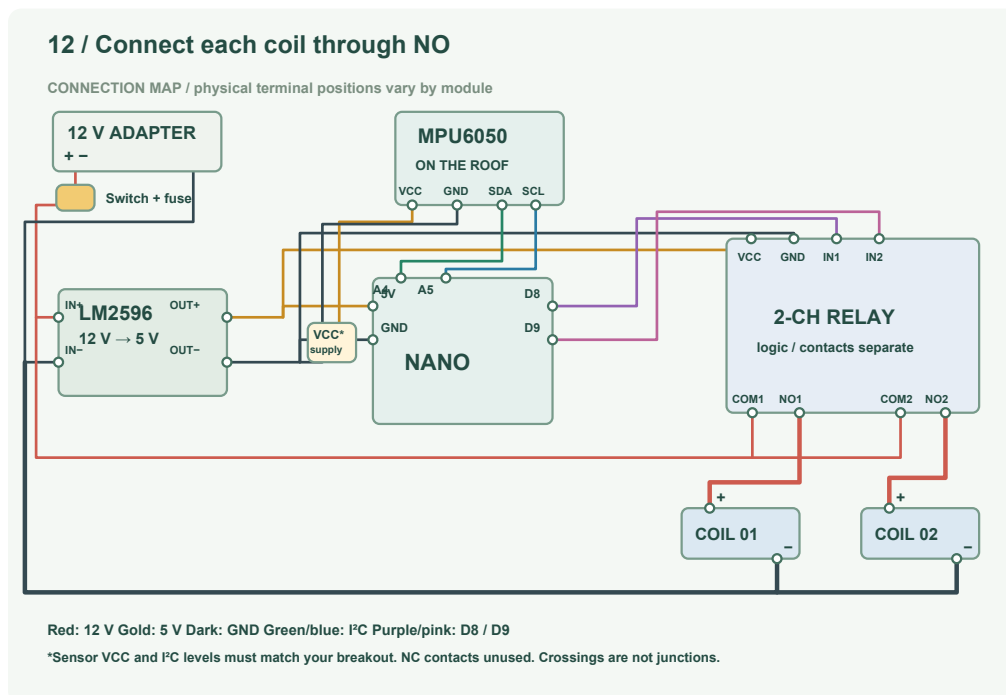
What to do

Connect switched 12 V to channel 1 COM and channel 2 COM. Leave both NC terminals unused. Use a meter and the module markings to confirm terminal order; do not infer the order from the blue housing orientation.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 12 / Connect each coil through NO

Disconnect all power before wiring. Relay contact terminals COM/NO/NC are separate from the module's logic pins.



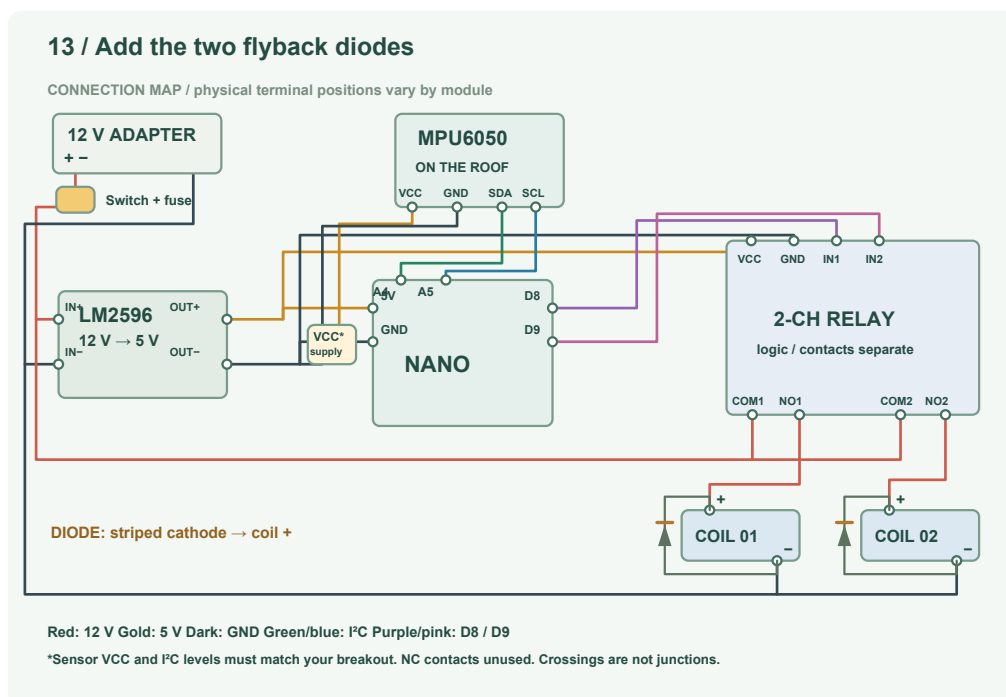
What to do

Connect channel 1 NO to solenoid 01 positive. Connect channel 2 NO to solenoid 02 positive. Return each solenoid negative directly to the common-ground distribution point using its own appropriately sized path. Keep these high-current returns separate from the sensor ground wire until the common point.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 13 / Add the two flyback diodes

Disconnect all power before wiring. Relay contact terminals COM/NO/NC are separate from the module's logic pins.



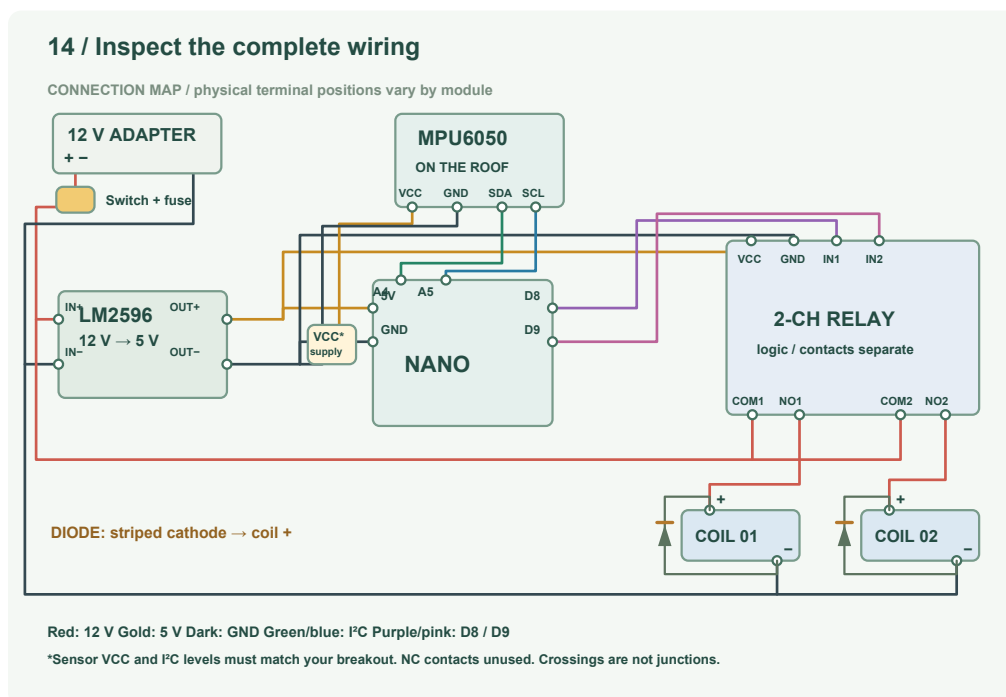
What to do

Place one suitable flyback diode across each coil, close to the solenoid terminals. Connect the striped cathode to coil positive (the relay NO side) and the anode to coil negative. Confirm polarity before applying power: a reversed diode can short the supply when the relay closes. Choose the diode for the coil current, reverse voltage and desired release behavior.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 14 / Inspect the complete wiring

Disconnect all power before wiring. Relay contact terminals COM/NO/NC are separate from the module's logic pins.



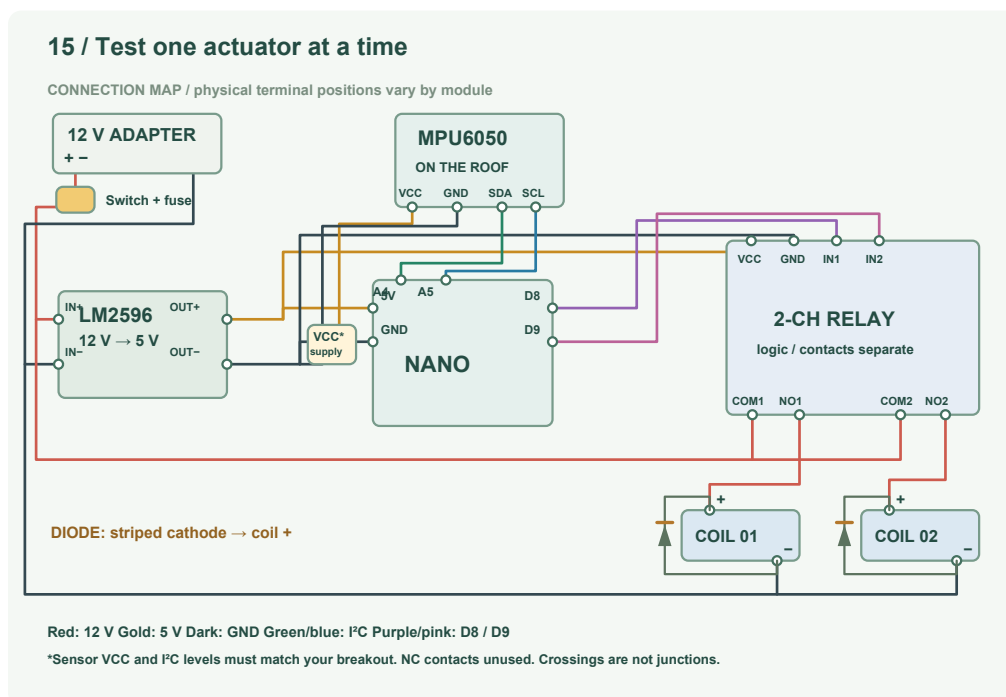
What to do

With power off, check for 12 V-to-ground shorts, loose strands, exposed conductors and mistaken relay terminals. Verify the buck still feeds only the control supply and the sensor receives its specified voltage. Confirm both solenoid positive wires are disconnected by inactive relay contacts. Confirm the physical switch removes the 12 V input.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 15 / Test one actuator at a time

Disconnect all power before wiring. Relay contact terminals COM/NO/NC are separate from the module's logic pins.



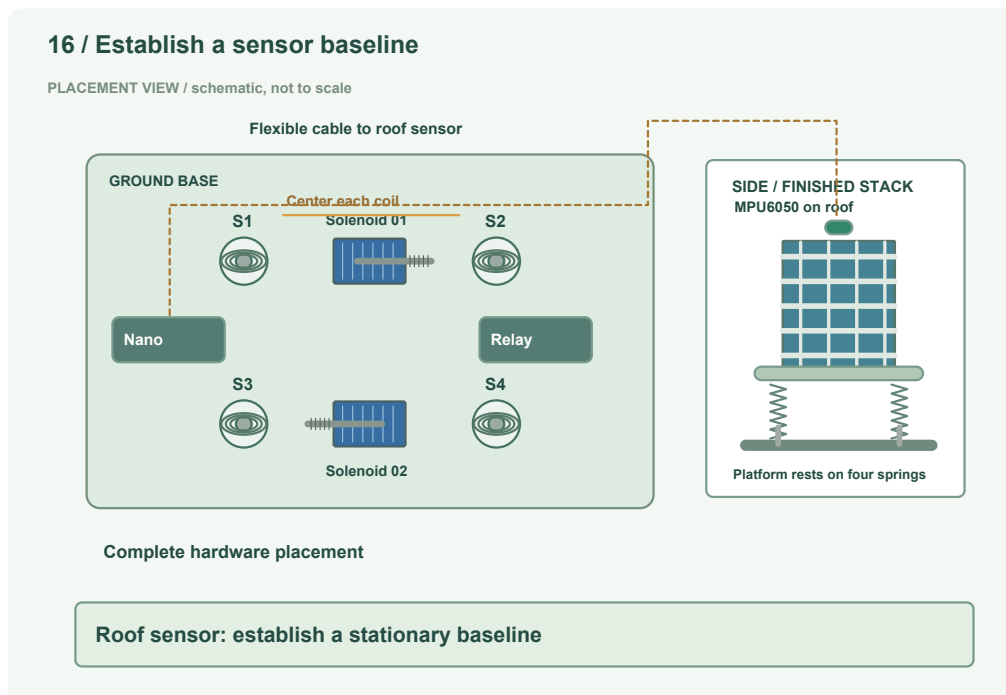
What to do

Keep the model clear and start with a brief pulse within the coil's duty rating. Observe the actual plunger direction and return action, then test the other channel. Check that no rod hits the platform, jams a spring or exceeds travel. Stop on binding or abnormal heating. Correct mounting or control polarity before enabling feedback.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 16 / Establish a sensor baseline

The website uses idealized forces. Real control must be tuned to the actual sensor, relays, coils and mechanical contact.



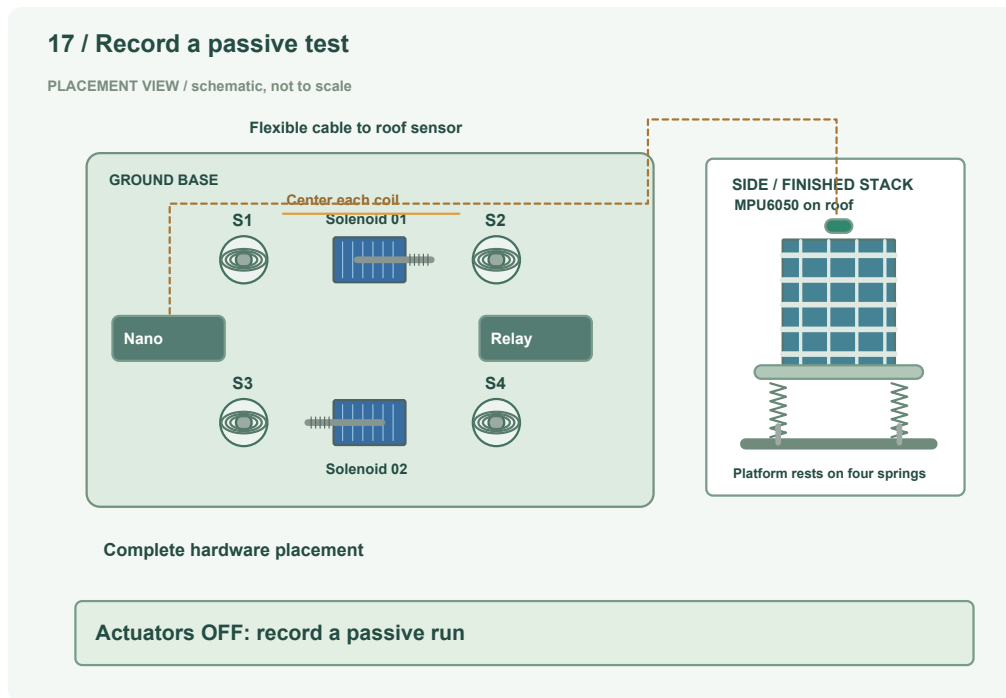
What to do

Keep the platform still, verify sensor communication, log acceleration and angular rate, and estimate stationary offsets. Mark which sensor axis aligns with the two solenoids. Confirm readings change in the expected direction when the platform is gently moved.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 17 / Record a passive test

The website uses idealized forces. Real control must be tuned to the actual sensor, relays, coils and mechanical contact.



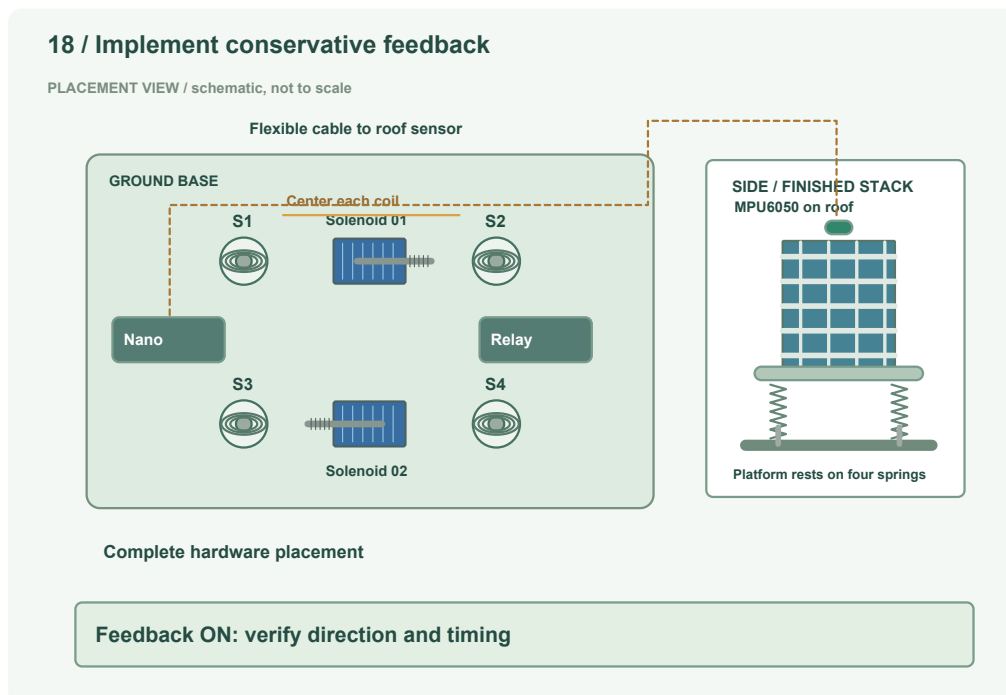
What to do

Leave the actuator outputs disabled and apply a small repeatable base movement. Record the input method, peak building response and settling time. Use the same model mass and initial conditions for subsequent comparisons. A roof sensor alone does not provide an independent ground-motion record.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 18 / Implement conservative feedback

The website uses idealized forces. Real control must be tuned to the actual sensor, relays, coils and mechanical contact.



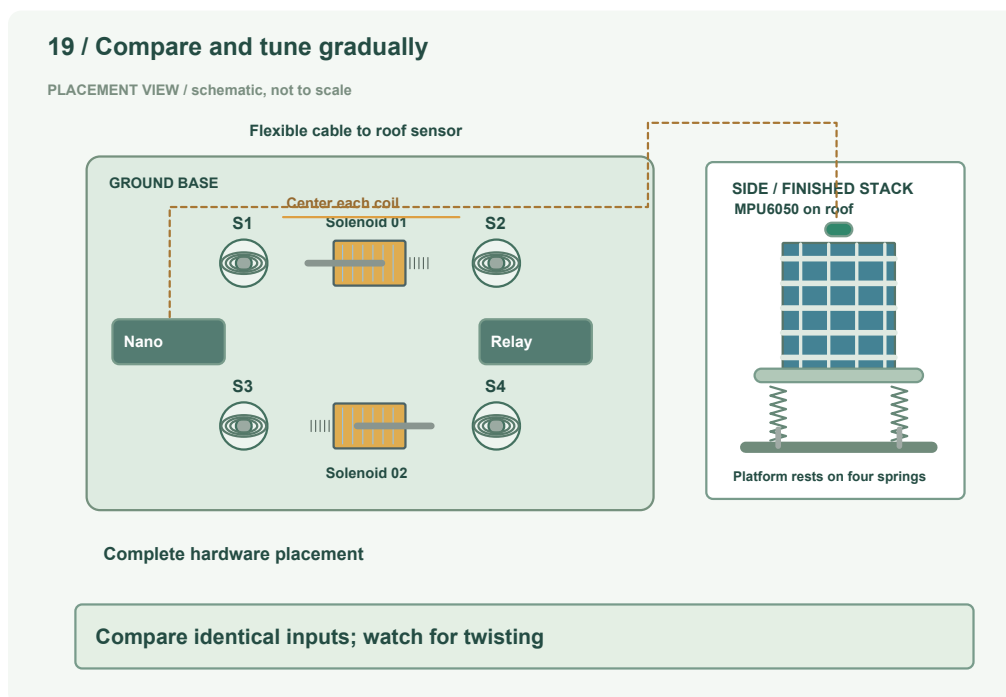
What to do

Use firmware that reads and filters motion, estimates the relevant movement, selects an opposing response, and releases each actuator appropriately. Include hysteresis, minimum switching intervals, maximum on-time, and default-OFF behavior for startup or sensor failure. Acceleration sign alone is not velocity direction; do not copy the continuous web damping force directly into relay switching.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 19 / Compare and tune gradually

The website uses idealized forces. Real control must be tuned to the actual sensor, relays, coils and mechanical contact.



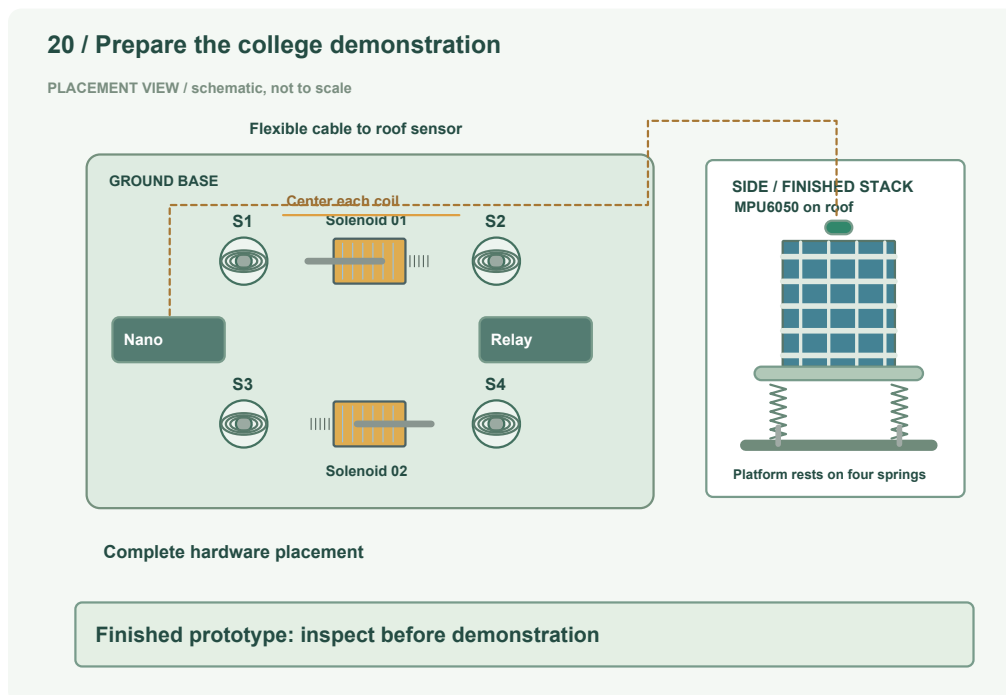
What to do

Repeat the same small base input with feedback enabled. Compare the traces and settling time against the passive run, and inspect for unwanted twisting caused by the offset actuators. If movement increases, stop and check direction, timing, sensor orientation and contact geometry. Tune one parameter at a time; do not increase excitation to hide a poor response.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

Step 20 / Prepare the college demonstration

The website uses idealized forces. Real control must be tuned to the actual sensor, relays, coils and mechanical contact.



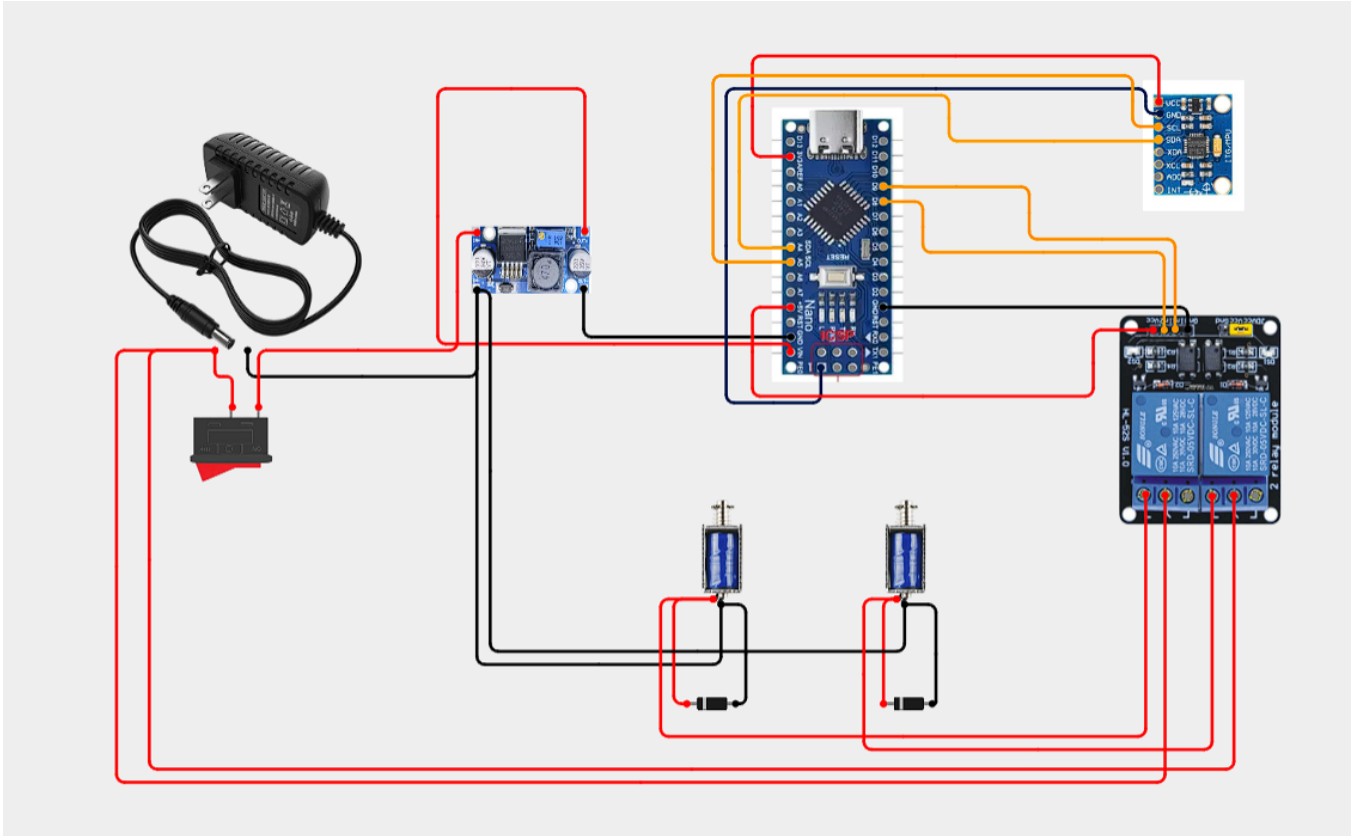
What to do

Secure the base, retain the springs, cover exposed power terminals, label all components and keep the power switch accessible. Demonstrate sensing first, then individual actuators, then passive versus controlled motion. Explain that the current mechanism is active solenoid response, not a proven eddy-current damping stage or a full-scale building protection system.

Placement and connection illustration, not a dimensioned fabrication drawing. Verify labels and ratings on your actual parts.

07 Wiring reference and notes

Original wiring image supplied with the project. Use the explicit pin table in section 04 when verifying connections; this image is a visual reference and does not specify mechanical mounting.



Details to settle before fabrication

Retain 1/2 inch (12.7 mm) screw insertion into each spring; select screw diameter and total length, spring free length, rate and retained end seats; record building mass and center of mass; confirm solenoid push/pull type, stroke, force and duty cycle; determine bracket and spring-contact geometry; verify control polarity and emergency power-off behavior.

Sources and revision history

Dynamics: OpenSees base-excitation equation. Actuation: NPTEL electrical and mechanical actuator dynamics. Model assumptions and constants are documented in the website physics panel.

[1] TDK InvenSense, MPU-6000 / MPU-6050 Product Specification. Sensor architecture and measurement principles. Official MPU-6050 product specification

[2] Arduino, classic Arduino Nano product documentation. I2C pin assignment. Official Arduino Nano documentation

Primary project inputs: owner-supplied detailed description, wiring image, original SEDAR PDF and subsequent four-spring / compact short-stroke push-pull / 1/2-inch insertion / roof-sensor clarifications. Revision 03 adds SI-unit dynamics and an illustrated 20-step build sequence. Original PDF retained separately as SEDAR-original-reference.pdf. Proposed brackets and collars are visualization assumptions, not owner-confirmed hardware.